PHATTHARAMON INWAN (KUNMING)

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PROFILE SUMMARY

A third-year undergraduate student in Nanomaterials Engineering with a strong academic foundation and a current interest in medical innovation, particularly biosensors, lab-on-a-chip systems, nanomaterials, and healthcare technologies. Has experience participating in sensor innovation competitions, earning multiple awards. Through these experiences, has developed strong teamwork skills, effective time management, and the ability to solve problems under time constraints, along with adaptability and resilience. Currently seeking an internship opportunity to further develop practical skills and apply academic knowledge toward future career growth in related fields.

INTERNSHIP EXPERIENCE

Nanotechnology Intern - Responsive Nanomaterials (RNM) Research Team, NSTDA — Apr 2026 - Jun 2026

- Contributed to the development of multiple microneedle-based technologies through extensive literature review, scientific analysis, and innovation-driven research. Experienced in evaluating research gaps, reviewing high-impact publications, and translating scientific findings into practical improvements for existing technologies. Proficient in research data acquisition, academic database searching, and evidence-based innovation development, with hands-on exposure to laboratory instruments and experimental methodologies.

PROJECT

Microguard — Dec 2025 - Present

- MicroGuard is an innovative system for detecting and capturing microplastics in water by integrating an eco-friendly olive oil–based separation method, nanoporous carbon adsorption, and AI Vision. The system achieves up to 99% microplastic recovery efficiency, providing a fast, accurate, and cost-effective solution for water quality monitoring and environmental protection.
- **Finalist (Top 11 Teams Worldwide)** at World Federation of Engineering Organizations (WFEO) Hackathon 2026
- **Selected as 1 of 111 Research Projects** for Poster Presentation at Thailand Research Expo & Symposium 2026.

Smart eyewear add-on — Sep 2025 - Jan 2026

- Smart eyewear add-on is a wearable health monitoring system for older adults that uses nanosensors, AI, and Fuzzy Logic to continuously monitor vital signs and environmental conditions. The system detects heat stroke risk in real time and instantly alerts users and caregivers through a smartwatch or mobile application, supporting preventive healthcare and independent living.
- **Finalist (Top 10 Teams)** at the International Conference “Hyper Interdisciplinary Conference (HIC) Thailand 2026
- **First Prize Winner (1st)** in Smart Materials Startup Showcase 2025

Easy Commu — Nov 2025

- Easy Commu is an AI-based sign language translation system that converts hand gestures into text in real time using a camera combined with artificial intelligence and machine learning (AI/ML) techniques for gesture recognition. The system is built using a YOLO-based object detection model, integrated with Python and OpenCV for real-time image processing. Curated and augmented datasets are used to improve recognition accuracy and robustness. Easy Commu helps bridge the communication gap between hearing and hearing-impaired communities, promoting social inclusion, equality, and sustainable accessibility in everyday communication.
- **2nd Runner-Up & Most Creative Award** in the BJM MediaX (Thammasat University) : AI Innovation Hackathon

EXPERIENCE

Attended the TMEC–Bruker Metrology Innovation Workshop 2025 at TMEC — Nov 2025

- Participated in the TMEC–Bruker Metrology Innovation Workshop 2025, gaining hands-on experience with advanced semiconductor metrology techniques, including optical microscopy, surface profilometry, and AFM. The cleanroom tour provided valuable insight into semiconductor manufacturing, nanomaterials characterization, and microfabrication processes, strengthening my understanding of industrial and research applications.

EDUCATION

King Mongkut's Institute of Technology Ladkrabang — July 2024 - Present

Bachelor of Nanomaterial Engineering

- Awarded the **First-Class Academic Excellence Scholarship** for maintaining the highest academic rank (1st) throughout Years 1–2 among 100+ students in Nanomaterials Engineering at KMITL.

Ramkhamhaeng University — July 2025 - Present

Bachelor of Economics

ADDITIONAL SKILLS

Computer Basics : *Proficient* MS Office (Excel, Word, Power Point) Canva, Google (Docs, Sheets, Slide)

Technical Skills : *Intermediate* Circuit, Programming (C, HTML&CSS, JS), SolidWorks

Language : Thai (native), English (able to communicate in daily and work related situations)

Communication & Collaboration

- Collaborative team player with strong communication and interpersonal skills.
- Excellent time management, responsibility, and adaptability in fast-paced environments.
- Open to feedback and committed to continuous improvement.